

Weekly Report — Manual-Agent Benchmark: from v1 to v2.1

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This week we focused on improving the manual agent according to the v1 evaluation result: every problem the v1 benchmark surfaced was filed as a GitHub issue, fixed in its own PR with a measured before/after score, and merged only on that evidence.

1 Headline scores

Milestone	Mean overall (30 locked goldens)	Notes
v1 (2026-06-20)	0.590	Confounded: 19/30 runs hit the time budget; metric artifacts under-counted correct answers. Not directly comparable to v2+.
v2 (2026-07-12)	0.670	Clean re-baseline after the eval-harness fixes; 0 timeouts.
v2 re-scored under today's rubric	0.713	Same v2 transcripts, current grader — the fair baseline for the v2.1 delta.
v2.1 (2026-07-13)	0.777	This week's agent + grader fixes combined. Median 0.811, σ 0.195, 0 timeouts, mean wall 73 s.

Same-rubric agent improvement this week: 0.713 $\rightarrow$ 0.777 (+0.064). Raw journey since v1: 0.590 $\rightarrow$ 0.777. The agent beats the RAG floor (0.239 at v2) on every question type.

Per question type (v2 $\rightarrow$ v2.1)

Question type	v2	v2.1
lookup	0.805	0.839
procedural	0.567	0.741
cross-section	0.608	0.752
image-required	0.660	0.779
adversarial	0.711	0.774

2 Problems found $\rightarrow$ fixes shipped $\rightarrow$ measured effect

Problem (short)	Issue / PR	Measured effect
Agent ran out of time/iterations on 19/30 questions	#143, #144, #165 / PR #163, #166, #164	Timeouts 19/30 $\rightarrow$ 0/30; mean wall 66 s vs 240 s budget
Judge could not credit a correct adversarial refusal	#146 / PR #169	Adversarial declines now scored: answer_quality 0 $\rightarrow$ 0.75–0.95; generic decline A/B 0.25 $\rightarrow$ 0.45
“Must not omit X” counted as hallucination (double penalty)	#147 / PR #170	Omissions moved to recall side; RAG hallucination_penalty 0.71 $\rightarrow$ 0.82

Problem (short)	Issue / PR	Measured effect
Slug-spelling mismatches zeroed correct citations	#145 / PR #168	citation_quality 0.427 → 0.733 (v1→v2 component)
Adversarial entries got a free +0.20 score floor	#148 / PR #176	section_recall → N/A + weight renormalisation; floor removed
CJK-exact must_contain under-counted correct English answers	#149 / PR #174	Bilingual fact matching (fact_recall 0.336 → 0.678 v1→v2, with completions)
Golden tool-count expectations unrealistic	#150 / PR #172	trajectory_efficiency 0.363 → 0.820
Golden manual identity drifted from corpus labels	#151 / PR #173	UUID-only gate at promotion; drift class eliminated
Agent made redundant tool calls	#152 / PR #171	Mean tool calls 6.17 → 5.77
Free structural-floor weight terms inflated both systems	#153 / PR #175	Zero-content floor 0.40 → 0.30
Re-baseline needed after all rubric fixes	#155 / PR #177	v2 pinned: agent 0.670 vs RAG 0.239; thresholds re-pinned 0.4 / 0.1
Long procedures lost steps; near-misses became “manual doesn’t contain X”	#184 / PR #185	procedural 0.567 → 0.709; dtc-001 0.338 → 0.993 (smoke)
Real section title invisible to TOC (marker-pdf missed heading)	#186 / PR #187	procedural-005 0.365 → 0.988; image-005 0.382 → 0.918
claim_precision punished valid alternate-section citations	#192 / PR #199	claim_precision 0.294 → 0.660; overall +0.043 (exact offline re-score)
Multi-part questions half-answered; wrong-manual reads discovered	#194 / PR #196	cross-section 0.624 → 0.724 (3 verify rounds); deterministic manual-pinning guard added
Figure-caption headings shadowed real chapters	#195 / PR #198	前煞車 136-char stub → full 27 kB chapter; zero golden slugs lost
Judge cannot see figures the agent surfaced	#193 / PR #197 — HELD	Plumbing complete, no regressions; merge held because the score lift is unproven (confounds filed as #200—#202)

Supporting artifacts: every PR carries its verification report; raw run data committed under docs/eval-reports/ (phase6_rebaseline_v2_eval.json, harness24_wp1_manual_lane.json, phase6_v2_1_manual_lane.json).

3 Honest caveats on v2.1

- Per-entry run variance remains the dominant noise source ($\sigma \approx 0.19$): e.g. dtc-001 scored 0.99 in two verification runs but 0.37 in this one (navigation stochasticity at temperature 0.2). Milestone means are trustworthy; single entries are not.
- 3 entries below the 0.4 floor this run: cross-004 (the only golden that never names its vehicle, so the new pinning guard cannot engage — filed as #204), dtc-001 (variance, above), procedural-002 (stubborn navigation case).
- v1 → v2 numbers are not same-rubric comparable (the grader was itself part of what we fixed); the same-rubric weekly delta is the 0.713 → 0.777 figure.

4 Next steps

- Expert review of the 15 OBD goldens at /goldens/obd (#191) — the OBD lane hit 100% pass after one tuning round (#117) but its goldens are still candidate-tier.
- Variance reduction + the held image-evidence PR: #200 (intermittent declines), #201 (judge parse fallback), #202 (vision descriptions at ingestion), then re-verify PR #197.
- #204 (cross-004 vehicle naming), #203 (pre-existing router test failure).
- Umbrella #116 continues: the next milestone eval after the above lands becomes v2.2.